

PAIR OF STRAIGHT LINES

Worksheet -1

PAIR OF STRAIGHT LINES

LEVEL-1

SINGLE CORRECT CHOICE TYPE:

1. Key : 1

Sol. Required equation is $(2x + y)(x + 2y) = 0 \Rightarrow 2x^2 + 5xy + 2y^2 = 0$

2. Key : 3

Sol. $3x^2 - 5xy + 2y^2 = 0 \Rightarrow 3x^2 - 3xy - 2xy + 2y^2 = 0 \Rightarrow 3x(x - y) - 2y(x - y) = 0$
 $\Rightarrow (x - y)(3x - 2y) = 0; x - y = 0; 3x - 2y = 0$

3. Key : 3

Sol. $\cos \theta = \frac{a+b}{\sqrt{(a-b)^2 + 4h^2}} = \frac{1+1}{\sqrt{(1-1)^2 + 16}} = \frac{2}{4} = \frac{1}{2} \Rightarrow \cos \theta = \frac{1}{2} \Rightarrow \theta = \frac{\pi}{3}$

4. Key : 1

Sol. Sum of the slopes $= \frac{5}{1} = 5$

MULTI CORRECT CHOICE TYPE:

5. Key : 1, 3

Sol. $6x^2 - xy + 4cy^2 = (3x + 4y)(kx - 3y) = 3kx^2 - 9xy + 4kxy - 12y^2$
 $= 3kx^2 - xy[9 - 4k] - 12y^2; 3k = 6 \Rightarrow k = 2; -12 = 4c \Rightarrow c = -3$

REASONING TYPE:

6. Key : 1

Sol. $\cos \theta = \frac{a+b}{\sqrt{(a-b)^2 + 4h^2}}$

COMPREHENSION TYPE:

7. Key : 4

Sol. $x^2 + 3xy - 4y^2 = 0; \cos \theta = \frac{a+b}{\sqrt{(a-b)^2 + 4h^2}} = \frac{1-4}{\sqrt{(1+4)^2 + 4\left(\frac{3}{2}\right)^2}} = \frac{-3}{\sqrt{25+9}} = \frac{-3}{\sqrt{34}}$

8. Key : 3

$$\text{Sol. } 2x^2 - 7xy + 3y^2 = 0; \quad \cos \theta = \frac{a+b}{\sqrt{(a-b)^2 + 4h^2}} = \frac{5}{\sqrt{1+49}} = \frac{5}{\sqrt{50}} = \frac{5}{5\sqrt{2}} = \frac{1}{\sqrt{2}}$$

$$\cos \theta = \frac{1}{\sqrt{2}} \Rightarrow \theta = \frac{\pi}{4}$$

9. Key : 2

$$\text{Sol. } 3x^2 + 11xy + 3y^2 = 0; \quad \cos \theta = \frac{9}{\sqrt{9+121}} = \frac{9}{\sqrt{130}} \Rightarrow \tan \theta = \frac{7}{9}$$

MATRIX MATCH TYPE:

10. Key: 1) $\rightarrow$ q; 2) $\rightarrow$ r; 3) $\rightarrow$ s; 4) $\rightarrow$ p

$$\text{a) } ax^2 + 2hxy + by^2 = k(y - m_1x)(y - m_2x) = k[y^2 - m_2xy - m_1xy + m_1m_2x^2]$$

$$\Rightarrow b\left[\frac{a}{b}x^2 + \frac{2h}{b}xy + y^2\right] = k[m_1m_2x^2 - xy(m_1 + m_2) + y^2]$$

$$\text{ii) } \frac{1}{m_1} + \frac{1}{m_2} = \frac{m_1 + m_2}{m_1m_2} = \frac{-2h}{\frac{a}{b}} = \frac{-2h}{a}$$

$$\text{iii) } m_1m_2 = \frac{a}{b}; \quad m_1 + m_2 = -\frac{2h}{b}$$

$$\text{iv) } (m_1 - m_2)^2 = (m_1 + m_2)^2 - 4m_1m_2 = \frac{4h^2}{b^2} - \frac{4a}{b} = 4\left[\frac{h^2 - ab}{b^2}\right]$$

$$\Rightarrow m_1 - m_2 = \frac{2\sqrt{h^2 - ab}}{b}$$

LEVEL-2

SINGLE CORRECT CHOICE TYPE:

11. Key : 3

Sol. Sum of the slopes = 4 (Product of the slopes)

$$\Rightarrow \frac{2c}{-7} = 4 \times \frac{1}{-7} \Rightarrow 2c = 4 \Rightarrow c = 2$$

12. Key : 1, 2, 3

$$\text{Sol. } 3x^2 - 8xy - 3y^2 = 0; \quad a = 3, \quad b = -3, \quad h = -4$$

$$m_1 + m_2 = \frac{-2h}{b} = \frac{8}{-3} = -\frac{8}{3}; \quad m_1 \times m_2 = \frac{a}{b} = -1$$

$$m_1 - m_2 = \frac{2\sqrt{h^2 - ab}}{b} = \frac{2\sqrt{16+9}}{-3} = \frac{10}{-3} = \frac{-10}{3}$$

LEVEL-3**SINGLE CORRECT CHOICE TYPE:**

13. Key : 1

$$\text{Sol. } x = \frac{-2y \cos \theta \pm \sqrt{4yz}}{2} = \frac{1}{2} \left[-2y \cos \theta \pm 2y \sqrt{\cos^2 \theta - 1} \right]$$

$$= -y \cos \theta \pm y \cot \theta; \quad x = -y (\cos \theta \pm \cot \theta)$$

$$\therefore \text{ The two lines are } x + y [\cos \theta + \cot \theta] = 0 \text{ and } x + y [\cos \theta - \cot \theta] = 0$$

MATRIX MATCH TYPE:14. Key: 1) $\rightarrow p$; 2) $\rightarrow q$; 3) $\rightarrow r$; 4) $\rightarrow s$

$$\text{Sol. 1) } 2x^2 + 5xy + 2y^2 = 0; \quad h = \frac{5}{2} \quad a = 2, \quad h = \frac{5}{2}, \quad b = 2$$

$$\tan \theta = \frac{2\sqrt{h^2 - ab}}{a + b} = \frac{2\sqrt{\frac{25}{4} - 4}}{4} = \frac{2\sqrt{25 - 16}}{4} = \frac{3}{4}$$

$$2) \quad 2x^2 - 5xy + 2y^2 = 0 \quad a = b = 2, \quad h = \frac{5}{2}$$

$$\cos \theta = \frac{a + b}{\sqrt{(a - b)^2 + 4h^2}} = \frac{4}{\sqrt{0 + 25}} = \frac{4}{5}$$

$$3) \quad xy + y^2 = 0; \quad a = 0, \quad b = 1, \quad h = \frac{1}{2}$$

$$\cos \theta = \frac{a + b}{\sqrt{(a - b)^2 + 4h^2}} \Rightarrow \cos \theta = \frac{1}{\sqrt{1+1}} = \frac{1}{\sqrt{2}} \Rightarrow \sin \theta = \sqrt{1 - \cos^2 \theta} = \frac{1}{\sqrt{2}}$$

$$4) \quad x^2 - 7xy + 12y^2 = 0 \quad a = 1, \quad b = 12, \quad h = \frac{-7}{2}$$

$$\sin \theta = \sqrt{1 - \cos^2 \theta} = \sqrt{1 - \frac{169}{170}} = \frac{1}{\sqrt{170}}$$

$$\therefore \frac{2 \cos \theta + 3 \sin \theta}{4 \sin \theta + 5 \cos \theta} = \frac{\frac{36}{\sqrt{170}} + \frac{3}{\sqrt{170}}}{\frac{4}{\sqrt{170}} + \frac{65}{\sqrt{170}}} = \frac{39}{69}$$

LEVEL-4**SINGLE CORRECT CHOICE TYPE:**

15. Key : 1

Sol. Required equation is $[y - \tan 30^\circ \times x][y - \tan 60^\circ \times x] = 0$

$$\Rightarrow \left[y - \frac{x}{\sqrt{3}} \right] [y - \sqrt{3} x] = 0 \Rightarrow y^2 - \sqrt{3} \times xy - \frac{xy}{\sqrt{3}} + x^2 = 0 \Rightarrow (x^2 + y^2)(\sqrt{3}) = 4xy$$

REASONING TYPE:

16. Key : 3

LEVEL-5**SINGLE CORRECT CHOICE TYPE:**

17. Key : 2

COMPREHENSION TYPE:

18. Key : 1

Sol. $A - B = 0$ is $2x^2 - 20xy + 8y^2 - (-2x^2 + 4xy - 3y^2)$

$$2x^2 - 20xy + 8y^2 + 2x^2 - 4xy + 3y^2 = 4x^2 - 24xy + 11y^2$$

$$4x^2 - 2xy - 22xy + 11y^2 = 2x(2x - y) - 11y(2x - y) = (2x - y)(2x - 11y)$$

19. Key : 3

Sol. $ab(x^2 - y^2) + (a^2 - b^2)xy = 0$; $abx^2 - b^2xy + a^2xy - aby^2 = 0$

$$bx(ax - by) + ay(ax - by) = 0; \quad (ax - by)(bx + ay) = 0$$

20. Key : 3

Sol. $x^2 + 2xy \sec \theta + y^2 = 0$; $y^2 + 2xy \sec \theta + x^2 = 0$

$$y = \frac{-2x \sec \theta \pm \sqrt{4x^2 \sec^2 \theta - 4x^2}}{2}; \quad y = -x \sec \theta \pm x \sqrt{\sec^2 \theta - 1}$$

$$y = -x \sec \theta \pm x \tan \theta$$

$$y = (-x \sec \theta + x \tan \theta) \text{ or } (-x \sec \theta - x \tan \theta)$$

$$y = x(\tan \theta - \sec \theta) \text{ and } y = x(-\sec \theta - \tan \theta)$$

Worksheet -2**PAIR OF STRAIGHT LINES**

LEVEL-1**SINGLE CORRECT CHOICE TYPE:**

1. Key : 2

Sol. Angle bisectors are $x \pm y = 0 \Rightarrow x + y = 0, x - y = 0$

2. Key : 1

Sol. Angle bisectors are $\frac{x+y+1}{\sqrt{2}} \pm \frac{x-y+1}{\sqrt{2}} = 0 \Rightarrow x+1=0, y=0$

3. Key : 4

Sol. Given lines are $3x - 4y + 7 = 0$ and $12x + 5y - 2 = 0$;

$$a_1a_2 + b_1b_2 = 3(12) + 4(5) > 0$$

$$\therefore \text{The acute angle bisector is } \frac{3x - 4y + 7}{\sqrt{9+16}} + \frac{12x + 5y - 2}{\sqrt{144+25}} = 0$$

$$\Rightarrow 13(3x - 4y + 7) + 5(12x + 5y - 2) = 0 \Rightarrow 99x - 27y + 81 = 0 \Rightarrow 11x - 3y + 9 = 0$$

$$4. \text{ Key : 4} \quad \text{Sol. Area} = \frac{n^2 \sqrt{h^2 - ab}}{|am^2 - 2hlm + bl^2|} = \frac{(-1)^2 \sqrt{4-1}}{|1-4+1|} = \frac{\sqrt{3}}{2} \text{sq.units}$$

MULTI CORRECT CHOICE TYPE:

5. Key : 1, 3

Sol. Equation to the pair of angle bisectors of $x^2 - 2axy - y^2 = 0$ is

$$-a(x^2 - y^2) - (1+1)xy = 0 \Rightarrow ax^2 + 2xy - ay^2 = 0$$

$$\therefore \frac{a}{1} = \frac{2}{-2b} = \frac{-a}{-1} \Rightarrow a = \frac{-2}{2b} \Rightarrow ab = -1$$

6. Key : 1

$$\text{Sol. Area} = \frac{16\sqrt{100-84}}{|7(2)^2 + 20(2)(-3) + 12(-3)^2|} = \frac{64}{28-120+108} = \frac{64}{16} = 4$$

COMPREHENSION TYPE:

7. Key : 1

$$\text{Sol. Angle bisectors are } \Rightarrow \frac{3x - 4y + 2}{5} \pm \frac{12x - 5y + 2}{13} = 0$$

$$\Rightarrow 5(12x - 5y + 2) \pm 13(3x - 4y + 2) = 0 \Rightarrow 99x - 77y + 36 = 0, 21x + 27y - 16 = 0$$

8. Key : 1

Sol. Pair of angle bisectors is $\frac{x^2 - y^2}{2 - 1} = \frac{xy}{\frac{-3}{2}} \Rightarrow \frac{x^2 - y^2}{1} = \frac{-2xy}{3} \Rightarrow 3x^2 + 2xy - 3y^2 = 0$

9. Key : 1

Sol. $\frac{-3}{2}(x^2 - y^2) - (2 - 1)xy = 0 \Rightarrow 3x^2 + 2xy - 3y^2 = 0$

MATRIX MATCH TYPE:

10. Key: 1) $\rightarrow r$; 2) $\rightarrow s$; 3) $\rightarrow p$; 4) $\rightarrow q$

Sol. 1) $x^2 + 4xy + y^2 = 0$, $x + y = 1$;
 $a = 1$, $h = 2$, $b = 1$, $l = 1$, $m = 1$, $n = -1$

$$\text{Area} = \frac{|\sqrt{4-1}|}{|1-2(2)(1)+1|} = \frac{\sqrt{3}}{2}$$

2) $x^2 - 3xy + y^2 = 0$, $x + y + 1 = 0$; $a = 1$, $b = 2$, $h = \frac{-3}{2}$, $l = m = n = 1$

$$\text{Area} = \frac{1\sqrt{\frac{9}{4}-1}}{|1+3+1|} = \frac{\frac{\sqrt{5}}{2}}{5} = \frac{\sqrt{5}}{10} = \frac{1}{2\sqrt{5}}$$

3) $12x^2 - 20xy + 7y^2 = 0$; $2x - 3y + 4 = 0$
 $a = 12$, $b = 7$, $h = -10$, $l = 2$, $m = -3$, $n = 4$

$$\text{Area} = \frac{16\sqrt{100-84}}{|108-2(-10)(2)(-3)+7(4)|} = \frac{16\sqrt{16}}{|108-120+28|} = \frac{64}{16} = 4$$

4) $(3x + 4y)^2 - 3(3y - 4x)^2 = 0$, $3x + 4y + 5 = 0$

$$9x^2 + 16y^2 + 24xy - 3[9y^2 + 16x^2 - 24xy] = 0$$

$$9x^2 + 16y^2 + 24xy - 27y^2 - 48x^2 + 72xy = 0;$$

$$39x^2 - 96xy + 11y^2 = 0; \quad 3x + 4y + 5 = 0$$

$$a = 39, \quad b = -48, \quad h = 11, \quad l = 11, \quad m = 4, \quad n = 5$$

$$\text{Area} = \frac{25\sqrt{(-48)^2 - 39(11)}}{[39(16) - 2(-48)(3)(4) + 11(9)]}$$

$$= \frac{25\sqrt{2304-429}}{|624+1152+99|} = \frac{25\sqrt{1875}}{1875} = \frac{25}{\sqrt{1875}} = \frac{25}{25\sqrt{3}} = \frac{1}{\sqrt{3}}$$

LEVEL-2**SINGLE CORRECT CHOICE TYPE:**

11. Key : 1

Sol. Comparing $y = x + c$ with $lx + my + n = 0$

$$\text{We get } l = 1, m = -1, n = c; \quad \text{Area} = \frac{c^2 \sqrt{h^2 - ab}}{|a(1)^2 - 2h(1)(-1) + b(1)^2|} = \frac{c^2 \sqrt{h^2 - ab}}{|a + 2h + b|}$$

MATRIX MATCH TYPE:12. Key: 1) $\rightarrow q$; 2) $\rightarrow r$; 3) $\rightarrow s$; 4) $\rightarrow p$

$$\text{Sol. 1) } 3x^2 - 5xy + 2y^2 = 0; \quad a = 3, \quad h = \frac{-5}{2}, \quad b = 2$$

 $\therefore$ The equation to the pair of bisectors of angles between the pair of lines

$$ax^2 + 2hxy + by^2 = 0 \text{ is } h(x^2 - y^2) = (a - b)xy$$

$$\therefore \frac{-5}{2}(x^2 - y^2) = (3 - 2)xy \Rightarrow \frac{-5}{2}(x^2 - y^2) = 1xy$$

$$\Rightarrow -5x^2 + 5y^2 = 2xy \Rightarrow 5x^2 + 2xy - 5y^2 = 0$$

$$2) \quad x^2 + 4xy + y^2 = 0; \quad a = 1, \quad b = 1, \quad h = 2$$

$$\Rightarrow h(x^2 - y^2) = (a - b)xy \Rightarrow 2(x^2 - y^2) = (1 - 1)xy$$

$$\Rightarrow 2(x^2 - y^2) = 0 \Rightarrow x^2 - y^2 = 0$$

$$3) \quad 2x^2 + 6xy + y^2 = 0; \quad a = 2, \quad h = 3, \quad b = 1$$

$$\Rightarrow h(x^2 - y^2) = (a - b)xy \Rightarrow 3(x^2 - y^2) = (2 - 1)xy$$

$$\Rightarrow 3x^2 - 3y^2 - xy = 0 \Rightarrow 3x^2 - xy - 3y^2 = 0$$

$$4) \quad 6x^2 + 5xy - 6y^2 = 0; \quad a = 6, \quad b = -6, \quad h = \frac{5}{2}$$

$$h(x^2 - y^2) = (a - b)xy \Rightarrow \frac{5}{2}(x^2 - y^2) = (6 + 6)xy$$

$$5x^2 - 5y^2 = (12xy) \times 2 \Rightarrow 5x^2 - 24xy - 5y^2 = 0$$

LEVEL-3

SINGLE CORRECT CHOICE TYPE:

13. Key : 2

$$\text{Sol. } 3x^2 + xy + by^2 = 0; \quad a = 3, \quad h = \frac{1}{2}, \quad b = b$$

$$h(x^2 - y^2) = (a - b)xy; \quad \frac{1}{2}(x^2 - y^2) = (3 - b)xy$$

$$x^2 - y^2 = 6xy - 2bxy; \quad x^2 - xy[6 - 2b] - y^2 = 0$$

$$6 - 2b = 14; \quad -2b = 8 \quad b = -4$$

14. Key : 1

$$\text{Sol. } 3x^2 - 2xy - 8y^2 = 0 \text{ and Area} = 5, \quad 2x + y - k = 0$$

$$a = 3, \quad h = -1, \quad b = -8, \quad l = 2, \quad m = 1, \quad n = -k$$

$$5 = \frac{k^2 \sqrt{1 + 24}}{|3 - 2(-1)(2)(1) - 8(2)^2|} \Rightarrow 5 = \frac{5k^2}{|3 + 4 - 32|}; \quad k^2 = 25; \quad k = 5$$

REASONING TYPE:

15. Key : 4

$$\text{Sol. The equation of AB, CA. CA are } 3x + 4y - 6 = 0 \quad \dots\dots\dots(1)$$

$$12x - 5y - 3 = 0 \quad \dots\dots\dots(2) \quad 4x - 3y + 12 = 0 \quad \dots\dots\dots(3)$$

By solving (1), (2) we get coordinates of B as $\left(\frac{2}{3}, 1\right)$

By solving (2), (3) we get coordinates of C as $\left(\frac{69}{16}, \frac{39}{4}\right)$

Now the equation of the bisectors of $\angle A$ between (1) and (2) are

$$\frac{3x + 4y - 6}{\sqrt{(3)^2 + (4)^2}} = \frac{\pm 4x - 3y + 12}{\sqrt{(4)^2 + (-3)^2}} \text{ i.e., } \frac{3x + 4y - 6}{5} = \pm \frac{4x - 3y + 12}{5}$$

$$3x + 4y - 6 = \pm(4x - 3y + 12) \quad \dots\dots\dots(4)$$

Taking the +ve sign on RHS of (4)

$$3x + 4y - 6 = -4x - 3y + 12 \rightarrow x - 7y + 18 = 0 \quad \dots\dots\dots(5)$$

Taking -ve sign on RHS of (4)

$$3x + 4y - 6 = 4x - 3y + 12 \rightarrow 7x + y + 6 = 0 \quad \dots\dots\dots(5)$$

In the LHS of II substituting $B\left(\frac{2}{3}, 1\right)$ we have $\frac{2}{3} - 7 + 18 = \frac{35}{3}$ which is +ve

In the LHS of (5) substituting the coordinates of $C\left(\frac{69}{16}, \frac{39}{4}\right)$

we have $\frac{69}{16} - \frac{273}{18} + 8 = \frac{735}{16}$, which is -ve

B and C lie on opposite side of (5)

Hence $x - y + 18 = 0$ is the required internal bisector $\angle A$

LEVEL-4

MULTI CORRECT CHOICE TYPE:

16. Key : 2

Sol. $y = |x| \Rightarrow y = \pm x \Rightarrow (y + x)(y - x) = 0 \Rightarrow x^2 - y^2 = 0$ and

$$a = 1, b = -1, g = f = h = 0; l = 1, m = 2, n = -2; \quad \text{Area} = \frac{4\sqrt{0+1}}{|4-0-1|} = \frac{4}{3} \text{ sq. units}$$

MATRIX MATCH TYPE:

17. Key: 1) $\rightarrow r$; 2) $\rightarrow s$; 3) $\rightarrow p$; 4) $\rightarrow q$

Sol. 1) $(x + y)^2 - (x - 3y)^2 = 0$ where $k = 1$

2) $(x + y)^2 - 3(x - y)^2 = 0$ where $k = 3$

3) $(x + y)^2 - \frac{1}{3}(x - y)^2 = 0$ where $k = \frac{1}{3}$

4) $(x + y)^2 (2 + \sqrt{3})^2 (x - y)^2 = 0$ where $\tan \alpha = 2 + \sqrt{3}$

LEVEL-5

MULTI CORRECT CHOICE TYPE:

18. Key : 1, 2

Sol. $y^2 + xy - 12x^2 = 0$; $y^2 + 4xy - 3xy - 12x^2 = 0$

$$y(y + 4x) - 3x(y + 4x) = 0; \quad (y - 3x)(y + 4x) = 0$$

Case - I : $y - 3x = 0 \Rightarrow y = 3x$; y value sub in $ax^2 + 2hxy + by^2 = 0$

$$a(3x)^2 + 2hx(3x) + b(3x)^2; \quad 9ax^2 + 6hx^2 + 9bx^2 = 0$$

$$a = -3(2h + 3b)$$

Case - II : $y + 4x = 0$; $y = -4x$

$$ax^2 + 2hxy + by^2 = 0; \quad a(-4x)^2 + 2h(x)(-4x) + b(-4x)^2 = 0$$

$$a16x^2 - 8hx^2 + 16bx^2 = 0; \quad 16a - 8h + 16b = 0$$

$$a = 8h - 16b; \quad a = 8(h - 2b) = 0$$

19. Key : 1, 4

Sol. Here $my(y - mx) + x(y - mx) = 0$; i.e., $(y - mx)(my + x) = 0$

So, the lines are $y = mx$ or $y = \frac{-1}{m}x$

Bisectors between the lines $xy = 0$ are $y = x$ or $y = -x$

$$\therefore m = 1 \quad \text{or} \quad m = -1$$

Worksheet -3

PAIR OF STRAIGHT LINES

1. Key : 1

Sol: $abc + 2fgh - af^2 - bg^2 - ch^2 = 0$

$$(k)(-3)(2) - 7(2)\left(\frac{7}{2}\right) - k \times \frac{49}{4} - (-3)(4) - \frac{2 \times 49}{4} = 0$$

$$\Rightarrow -6k - \frac{49}{4}k + 61 - \frac{49}{2} = 0; \Rightarrow -73k + 146 = 0; \Rightarrow 146 = 73k \Rightarrow k = 2$$

2. Key : 4

Sol: $xy + x + y + 1 = 0 \Rightarrow x(y + 1) + (y + 1) = 0$
 $\Rightarrow (y + 1)(x + 1) = 0$
 $\Rightarrow x = -1, y = -1$

$\therefore$ point of intersection = $(-1, -1)$

3. Key : 2

Sol: Distance between the parallel lines

$$= 2\sqrt{\frac{g^2 - ac}{a(a+b)}} = 2\sqrt{\frac{81 - 72}{90}} = 2\sqrt{\frac{9}{90}} = \frac{2}{\sqrt{10}}$$

4. Key : 2

Sol: $(1, k)$ is the point of intersection

$$\begin{aligned} \Rightarrow 2 + 5k + 3k^2 + 6 + 7k + 4 &= 0 & \Rightarrow 3k^2 + 12k + 12 &= 0 \\ & & \Rightarrow k^2 + 4k + 4 &= 0 \\ & & \Rightarrow (k + 2)^2 &= 0 \Rightarrow k = -2 \end{aligned}$$

5. Key : 1, 2

Sol: $4x^2 + 2kxy + y^2 + 6x + ky - 10 = 0$

$$a = 4, \quad b = 1, \quad g = 3, \quad c = -10$$

$$\text{Distance between parallel lines} = 2\sqrt{\frac{g^2 - ac}{a(a+b)}}$$

$$2\sqrt{\frac{9+40}{4(4+1)}} = 2\sqrt{\frac{49}{20}} = 2 \times \frac{7}{2\sqrt{5}} = \frac{7}{\sqrt{5}}$$

6. Key : 1

$$\text{Sol: St-I: } a = 1, \quad b = -1, \quad h = 0, \quad g = 0, \quad f = \frac{k}{2}, \quad c = -a^2$$

$$abc + 2gfh - ab^2 - bg^2 - ch^2 = 0$$

$$\Rightarrow 1(-1)(-a^2) + 2(0)\left(\frac{k}{2}\right)(0) - 1\left(\frac{k}{2}\right)^2 - (-1)(0)^2 - (-a^2)(0)^2 = 0$$

$$\Rightarrow a^2 + 0 - \frac{k^2}{4} - 0 - 0 = 0; \quad a^2 = \frac{k^2}{4} \Rightarrow k^2 = 4a^2 \Rightarrow k = \pm 2a$$

7. Key : 4

$$\text{Sol: } xy + 2x + 2y + 4 = 0 \Rightarrow x(y + 2) + 2(y + 2) = 0$$

$$\Rightarrow (x + 2)(y + 2) = 0 \Rightarrow x = y = -2$$

$\therefore$ point of intersection is $(-2, -2)$.

8. Key : 1

$$\text{Sol: Point of intersection} = \left(\frac{hf - bg}{ab - h^2}, \frac{gh - af}{ab - h^2} \right)$$

$$2x^2 + 5xy + 2y^2 - 17x - 19y + 35 = 0$$

$$a = 2, \quad h = \frac{5}{2}, \quad b = 2, \quad g = \frac{-17}{2}, \quad f = \frac{-19}{2}, \quad c = 35$$

$$\Rightarrow \left(\frac{\frac{5}{2}\left(\frac{-19}{2}\right) - 2\left(\frac{-17}{2}\right)}{4 - \frac{25}{4}}, \frac{\frac{-17}{2} \times \frac{5}{2} - 2\left(\frac{-19}{2}\right)}{4 - \frac{25}{4}} \right) \Rightarrow \left(\frac{-95 + 68}{16 - 25}, \frac{-85 + 76}{16 - 25} \right) = (3, 1)$$

9. Key : 4

$$\text{Sol: Point of intersection} = \left(\frac{\frac{-13}{2} \times \frac{41}{2} - (-7)(11)}{2(-7) - \left(\frac{-13}{2}\right)^2}, \frac{11\left(\frac{-13}{2}\right) - 2\left(\frac{41}{2}\right)}{2(-7) - \left(\frac{-13}{2}\right)^2} \right)$$

$$= \left(\frac{-533+308}{-56-169}, \frac{-286-164}{-56-169} \right) = (1, 2)$$

10. Key : $a \rightarrow 3, \quad b \rightarrow 1, \quad c \rightarrow 4, \quad d \rightarrow 2$

Sol: If the equation $ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$ represents a pair of straight line then

$$abc + 2fgh - af^2 - bg^2 - ch^2 = 0$$

a) If $a = 0 \Rightarrow 0 + 2fgh - 0 - bg^2 - ch^2 = 0$

$$\Rightarrow 2fgh = bg^2 + ch^2$$

b) If $b = 0 \Rightarrow 0 + 2fgh - af^2 - 0 - ch^2 = 0$

$$\Rightarrow 2fgh = af^2 + ch^2$$

c) If $c = 0 \Rightarrow 0 + 2fgh - af^2 - bg^2 - 0 = 0$

$$\Rightarrow 2fgh = af^2 + bg^2$$

d) If $f = 0 \Rightarrow abc + 0 - 0 - bg^2 - ch^2 = 0$

$$\Rightarrow abc = bg^2 + ch^2$$

11. Key : 1

Sol: $4x^2 + 2kxy + y^2 + 6x + ky - 10 = 0$

$$a = 4, \quad h = k, \quad b = 1, \quad g = 3, \quad f = \frac{k}{2}, \quad c = -10$$

$$\Rightarrow h^2 = ab \Rightarrow k^2 = 4 \Rightarrow bg^2 = af^2$$

$$\Rightarrow k = \pm 2 \Rightarrow 9 = k^2 \Rightarrow k = \pm 3$$

12. Key : 1, 2

Sol: $kx^2 - 10xy + 12ky^2 + 5x - 16y - 3 = 0$

$$\Rightarrow a = k, \quad h = -5, \quad b = 12k, \quad g = \frac{5}{2}, \quad f = -8, \quad c = -3$$

$$\Rightarrow h^2 = ab \Rightarrow (-5)^2 = k(12k) \Rightarrow 25 = 12k^2 \Rightarrow k^2 = \frac{25}{12} \Rightarrow k = \frac{\pm 5}{2\sqrt{3}}$$

13. Key : 3

Sol: Conceptual

14. Key : 1

Sol: Conceptual

15. Key : 2

Sol: $2xy + gx + fy + 5 = 0; \quad a = 0, \quad b = 0, \quad c = 5, \quad h = 1, \quad g = \frac{g}{2}, \quad f = \frac{f}{2}$

$$\therefore abc + 2fgh - af^2 - bg^2 - ch^2 = 0$$

$$\Rightarrow 0 + 2\left(\frac{f}{2}\right)\left(\frac{g}{2}\right)(1) - 0 - 0 - 5(1)^2 = 0 \Rightarrow \frac{2fg}{4} - 5 = 0 \Rightarrow fg = 10.$$

16. Key : 2

Sol: Conceptual

17. Key : $a \rightarrow 4, \quad b \rightarrow 3, \quad c \rightarrow 2, \quad d \rightarrow 1$

Sol: Conceptual

18. Key : 2

Sol: Conceptual

19. Key : $a \rightarrow 4$, $b \rightarrow 3$, $c \rightarrow 2$, $d \rightarrow 1$

Sol: Conceptual

20. Key : 2 Sol: Conceptual

Worksheet -4**PAIR OF STRAIGHT LINES****LEVEL-1****SINGLE CORRECT CHOICE TYPE:**

1. Key : 3
2. Key : 3
3. Key : 3

MULTI CORRECT CHOICE TYPE:

4. Key : 1, 4

Sol. Equation to the line can be rearranged as $\frac{x+2y}{k} = 1$

Homogenising the equation of the maine with the help at that of the line

$$x^2 \left(2 + \frac{2}{k} - \frac{1}{k^2} \right) + xy \left(-2 + \frac{4}{k} - \frac{1}{k} - \frac{4}{k^2} \right) + y^2 \left(3 - \frac{2}{k} - \frac{4}{k^2} \right) = 0.$$

The pair of lines will be perpendicular. If coefficient of x^2 + coefficient of $y^2 = 0$

$$2 + \frac{2}{k} - \frac{1}{k^2} + 3 - \frac{2}{k} - \frac{4}{k^2} = 0 \Rightarrow 5k^2 - 5 = 0 \rightarrow k^2 = 1 \rightarrow k = +1 \text{ or } k = -1$$

REASONING TYPE:

5. Key : 1

Sol. $x^2 + y^2 + 2xy + x + 3y + 1 = 0$ and $x + y = 1$

$$\Rightarrow x^2 + y^2 + 2xy + (x + 3y)(x + y) + 1(x + y)^2$$

$$\Rightarrow x^2 + y^2 + 2xy + x^2 + xy + 3xy + 3y^2 + x^2 + y^2 + 2xy = 0 \Rightarrow 3x^2 + 5y^2 + 8xy = 0$$

COMPREHENSION TYPE:

6. Key : 1

Sol. $(x - h)^2 + (y - k)^2 = c^2$; $x^2 + h^2 - 2xh + y^2 + k^2 - 2yk - c^2 = 0$

$$x^2 + h^2 + k^2 = 2yh + 2yh + c^2$$

7. Key : 1

Sol. In homogenising we get

$$x^2 + y^2 - 2xh \frac{(kx + hy)}{2hk} - 2yk \frac{(kx + hy)}{2hk} + (h^2 + k^2 - c^2) \left(\frac{kx + hy}{2hk} \right)^2 = 0$$

$$x^2 + y^2 - x \frac{(kx + hy)}{k} - y \frac{(kx + hy)}{h} + (h^2 + k^2 - c^2) \left(\frac{kx + hy}{2hk} \right)^2 = 0$$

8. Key : 4

MATRIX MATCH TYPE:

9. Key : 1-s; 2-p; 3-q; 4-r

LEVEL-2

SINGLE CORRECT CHOICE TYPE:

10. Key : 2

Sol. $x^2 + 2y^2 - 4 = 0$; $x^2 + 2y^2 - 4.1^2 = 0$; $x^2 + 2y^2 - 4(x + y)^2 = 0$

$$x^2 + 2y^2 - 4(x^2 + y^2 + 2xy) = 0; \quad x^2 + 2y^2 - 4x^2 - 4y^2 - 8xy = 0$$

$$-3x^2 - 2y^2 - 8xy = 0; \quad 3x^2 + 2y^2 + 8xy = 0$$

11. Key : 1

REASONING TYPE:

12. Key : 1

Sol. Rewriting $y + tx = 2at + a + 3$ as $y + tx = 1$, $2at + a + 3$

we have the equation of the pair of straight line joining the origin 0 to the

intersection points parabola as $y^2 - 4ax \frac{(y + tx)}{(2at + t^3)} = 0$.

i.e. $(2at + at^3)y^2 - 4axy - 4at^2x^2 = 0$. As the lines are mutually perpendicular coefficient of x^2 + coefficient of $y^2 = 0$.

$$-4at + 2at + at^3 = 0; \quad -2at + t^3 = 0; \quad at(t^2 - 2) = 0; \quad t = v, \pm\sqrt{2}$$

$t = 0$ is not queptable. the slopes of the possible lines AB $\pm\sqrt{2}$

LEVEL-3

SINGLE CORRECT CHOICE TYPE:

13. Key : 3

Sol. $x^2(c^2 - a^2m^2) - 2ma^2xy + y^2(c^2 - a^2) = 0$ as the angle is 90°

$$\text{coefficient of } x^2 + \text{coefficient of } y^2 = 0; \quad c^2 - a^2m^2 + 0^2 - a^2 = 0; \quad 2c^2 = a^2(1 + m^2)$$

MULTI CORRECT CHOICE TYPE:

14. Key : 1, 2, 3

Sol. Rewriting the equation of the given line $ax + by - 2ab = 0$; as $\frac{ax + by}{2ab} = 1$ and

the equation of the given curve is $x^2 + y^2 - 2(bx + ay) + (a^2 + b^2 - c^2) = 0$ on homogenising we get

$$x^2 + y^2 - 2(bx + ay) + \frac{(ax + by)^2}{2ab} + (a^2 + b^2 + c^2) \left(\frac{ax + by}{2ab} \right)^2 = 0$$

on simplification we get $(a^2x^2 + b^2y^2)(a^2 + b^2 - c^2) - 2ab(a^2 + b^2 + c^2)xy = 0$

As the two lines are perpendicular. The coefficient of x^2 + the coefficient of $y^2 = 0$

$$(a^2 + b^2 - c^2)(a^2 + b^2) = 0; \quad a^2 + b^2 - c^2 = 0; \quad a^2 + b^2 = c^2$$

LEVEL-4

SINGLE CORRECT CHOICE TYPE:

15. Key : 4

Sol. $x^2 + y^2 - 4 = 0; \quad x^2 + y^2 - 4\left(\frac{y-3x}{c}\right)^2 = 0$

$$x^2 + y^2 - 4\frac{(y^2 + 9x^2 - 6xy)}{c^2} = 0; \quad c^2x^2 + c^2y^2 - 4y^2 - 36x^2 + 24xy = 0 \times c^2$$

$$c^2x^2 - 36x^2 + c^2y^2 - 4y^2 + 24xy = 0; \quad x^2(c^2 - 36) + cy^2(c^2 - 4) + 24xy = 0$$

Put $c^2 = 20$; $x^2(20 - 36) + y^2(20 - 4) + 24xy = 0$

As coefficient of x^2 + coefficient of $y^2 = 0$; i.e., $-16 + 16 = 0$

$\therefore c^2 = 20$ satisfies the given condition

16. Key : 1-r; 2-p; c-q; d-t

Sol. Put $1 = 2x + 3y$; $x^2 + y^2 = 4(2x + 3y)^2$; $x^2 + y^2 = 4(4x^2 + 9y^2 + 12xy)$

$$10x^2 + 36y^2 + 48xy - x^2y^2 = 0$$

$$15x^2 + 35y^2 + 48xy = 0$$

$$a = 15 \quad b = 35 \quad c = 48$$

$$a + b + c = 15 + 35 + 48 = 98$$

LEVEL-5

COMPREHENSION TYPE:

17. Key : 2

18. Key : 3

Sol. $\frac{x^2}{a^2} + \frac{y^2}{b^2} = (1)^2 \rightarrow \frac{x^2}{a^2} + \frac{y^2}{b^2} = \left(\frac{lx + my}{-n}\right)^2 = \frac{x^2}{a^2} + \frac{y^2}{b^2} = \frac{l^2x^2 + m^2y^2 + 2lmxy}{x^2}$

$$\frac{n^2x^2}{a^2} + \frac{n^2y^2}{b^2} = l^2x^2 + m^2y^2 + 2lmxy = \frac{b^2x^2}{a^2} - l^2x^2 + \frac{n^2y^2}{b^2} - m^2y^2 - 2lmxy = 0$$

$$(Ax^2 + By^2 - 2hxy) = 0; \quad x^2 \frac{(h^2 - l^2)}{a^2} + y^2 \frac{(h^2 - m^2)}{b^2} - 2lmxy = 0$$

19. Key : 3

Sol. If they are coincident $H^2 - AB = 0$;

$$l^2m^2 = \left(\frac{h^2}{a^2} - l^2\right)\left(\frac{h^2}{b^2} - m^2\right);$$

$$l^2m^2 = \frac{h^4}{a^2b^2} - \frac{h^2m^2}{a^2} - \frac{l^2h^2}{b^2} + l^2m^2$$

